

Virtual Sensor-Guided Noise Diagnosis and Self-Correction Framework for Sensor Health Monitoring

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Abstract: Reliable sensor measurements are essential for engine control, combustion monitoring, and diagnostics, yet physical sensors are susceptible to measurement noise, bias, drift, and other faults that can degrade prediction accuracy. This work presents a machine learning–based virtual sensing framework for robust combustion monitoring using data from a heavy-duty diesel engine. A multi-target artificial neural network is developed to estimate key combustion variables, including intake air mass flow (MF_IA), NOx emissions, and start of combustion (SOC), from six measured engine signals. The virtual sensor is first evaluated under nominal operating conditions, demonstrating accurate prediction performance across a wide range of engine speeds and loads. Feature sensitivity analysis is then performed to identify the most influential input sensors and provide physical insight into the learned model. To investigate robustness, artificial sensor faults are injected into selected input signals, and the resulting prediction degradation is quantified. The proposed framework establishes a foundation for virtual sensor–guided sensor health monitoring by relating input sensor importance to model robustness under realistic fault conditions. Future work will extend this framework to residual-based fault diagnosis, sensor health assessment, and adaptive signal self-correction for intelligent engine monitoring systems.

Keywords: *Virtual Sensing, Machine Learning, Combustion Monitoring, Sensor Health Monitoring, Noise Diagnosis*

1. Introduction

Modern automotive engines rely on accurate sensing for combustion control, emissions reduction, and system diagnostics. However, physical sensors are susceptible to measurement noise, bias, drift, and other degradation mechanisms that reduce reliability over time. Virtual sensing provides a software-based alternative by estimating difficult-to-measure combustion variables from existing engine control unit (ECU) signals. In this work, a machine learning–based virtual sensor is developed to simultaneously estimate intake air mass flow (MF_IA), engine-out NOx emissions, and start of combustion (SOC). Beyond prediction accuracy, the proposed framework investigates how sensor faults propagate through the virtual sensor using sensitivity analysis and controlled noise injection, providing the foundation for future sensor health monitoring.

2. Methodology

2.1 Engine Dataset

Experimental data were collected from a 12-L six-cylinder heavy-duty diesel engine operating under 217 steady-state conditions covering a wide range of engine speeds and loads. Six ECU-accessible variables were used as model inputs: engine torque, ambient pressure, intake manifold pressure, intake manifold temperature, EGR rate, VTG position. The virtual sensor simultaneously predicts intake air mass flow (MF_IA), engine-out NOx, start of combustion (SOC). The dataset was randomly divided into 80% training and 20% testing subsets.

2.2 Machine Learning Virtual Sensor

A feed-forward artificial neural network with three hidden layers (64–32–16 neurons) was developed to learn the nonlinear relationship between engine operating conditions and combustion variables. A shared-representation architecture was adopted so that common physical dependencies among airflow, combustion phasing, and NOx formation could be learned simultaneously. Following model training, permutation-based feature sensitivity analysis was performed to quantify the influence of each input sensor on virtual sensor predictions.

3. Results and Discussion

The proposed ANN achieved accurate prediction of all three combustion variables across the testing dataset (Figure 1).

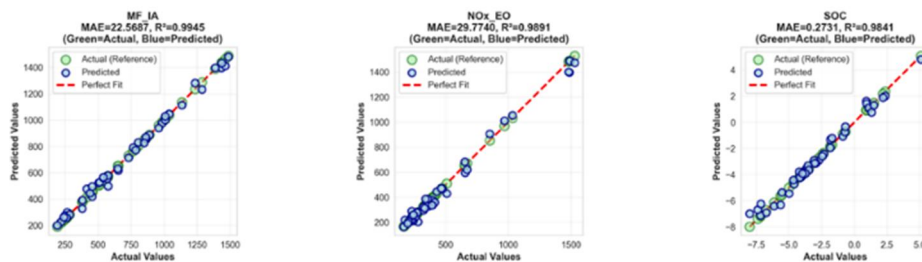


Figure 1 Predicted versus measured values for (a) intake air mass flow, (b) engine-out NOx concentration, and (c) start of combustion across the test dataset.

Table 1 Test-set performance summary

Target	MAE	R ²
MF_IA	22.6	0.995
NOx	29.8 ppm	0.989
SOC	0.27°C	0.984

Prediction accuracies reached $R^2 = 0.995$ for MF_IA, 0.989 for NOx, and 0.984 for SOC, demonstrating that the virtual sensor successfully captures the nonlinear relationships between engine operating conditions and combustion behavior. Intake manifold pressure and engine torque were the dominant inputs for airflow prediction, while EGR rate contributed most strongly to NOx estimation due to its influence on oxygen concentration and combustion temperature. These results indicate that the learned model reflects underlying combustion physics rather than purely statistical correlations.

4. Noise Injection and Robustness Evaluation

To evaluate virtual sensor robustness, representative operating conditions were selected using a quantile-median sampling strategy, in which the dataset was divided into low- and high-output regions and the sample closest to each group median was chosen. As a representative case, the MF_IA-High operating condition was selected, and 5% Gaussian noise was injected into the intake manifold pressure (P_IM) sensor, identified by sensitivity analysis as a critical input. A baseline sensor signal containing 2% Gaussian measurement noise was used to emulate normal operating conditions, while the corrupted signal was supplied to the trained ANN without retraining.

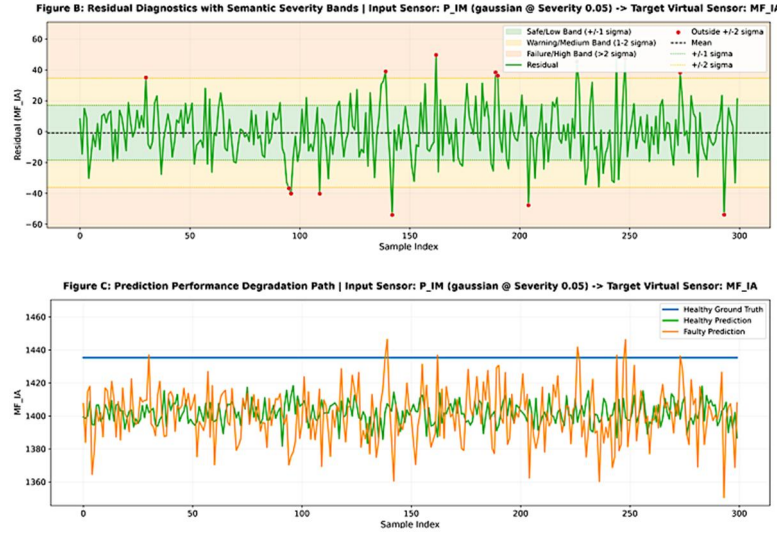


Figure 2 Representative robustness evaluation for the MF_IA-High operating condition. (a) Residual response after Gaussian noise injection into the P_IM sensor. (b) Comparison of healthy and faulty virtual sensor predictions.

Figure 2 shows the corresponding residual and prediction responses. The residual exhibits significantly larger high-frequency fluctuations and intermittent peaks after fault injection, indicating that independent measurement noise introduced into the physical sensor is not compensated by the virtual sensor prediction. Although the overall prediction trend is preserved, the corrupted input produces increased output variability together with a noticeable downward shift in the predicted MF_IA values. This behavior suggests that input noise propagates through the nonlinear neural network and introduces both random fluctuations and a systematic prediction bias. The observed degradation is consistent with the sensitivity analysis, confirming that faults affecting highly influential sensors have the greatest impact on virtual sensor performance.

These results demonstrate the feasibility of using virtual sensor residuals as indicators of sensor degradation and provide a foundation for future research on residual-based fault diagnosis, sensor health assessment, and adaptive signal self-correction.

5. Reference

- Alippi, C., Boracchi, G., and Roveri, M., IEEE Trans. Instrum. Meas., 2019, 68, 1456.
- Bhatt, A.N., and Shrivastava, N., Arch. Comput. Methods Eng., 2021, 29, 897.
- Qiu, Y., Yang, C., Zhang, J., Hametner, C., Deng, K., Zeng, T., and Lai, P., J. Power Sources, 2025, 655, 237975.
- Yap, W.K., and Karri, V., Appl. Energy, 2011, 88, 4505.